



Propensity Score Matching Analysis for the Patients of Unruptured Cerebral Aneurysm from a Post Hoc Analysis of a Nationwide Database in Japan

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■ BACKGROUND: This study aimed to compare the functional outcome at discharge for unruptured cerebral aneurysms (UCAs) between surgical clipping and endovascular coiling in total, nonelderly (<65 years), and elderly (≥65 years) patients by nonbiased analysis based on a national database in Japan.

■ METHODS: A total of 15,671 patients with UCA were registered in the Diagnosis Procedure Combination, the nationwide database, from 2010 to 2015 in Japan. The outcome of the Barthel Index (BI) at discharge was investigated, and propensity score–matched analysis was conducted in total, nonelderly, and elderly patient groups.

■ RESULTS: Propensity score–matched analysis found no significant difference for in-hospital mortality between the 2 treatment methods in the total and both age-groups. The rate of morbidity of BI <90 at discharge was higher after surgical clipping than after endovascular coiling in the total (4.9% vs. 3.9%; $P = 0.040$; risk difference, -1.0% ; 95% confidence interval, -3.6 to 2.3%) and the elderly age-group (8.1% vs. 5.0%; $P < 0.001$; risk difference, -3.1% ; -4.8% to 1.5%), however, no significant association between the 2 treatment methods (2.4% vs. 2.6%; $P = 0.67$; risk difference, 0.22% ; -0.79 to 1.22%) was found in the nonelderly group.

■ CONCLUSIONS: In elderly patients with UCA, a better outcome at discharge after endovascular coiling was

found. However, no significantly different functional outcome at discharge between surgical clipping and endovascular coiling for UCA in nonelderly patients was confirmed by propensity score–matched analysis from a nationwide database in Japan.

INTRODUCTION

Many asymptomatic unruptured cerebral aneurysms (UCAs) have recently been found by magnetic resonance imaging screening. The precise and unbiased results after surgical clipping and endovascular coiling were unclear in nonelderly and elderly patients with UCA; however, some patients should receive interventional treatment according to the individual risk of rupture. The number not only of surgical clipping but also of endovascular coiling has been increasing recently for elderly patients with UCA per 100,000 Medicare beneficiaries.¹ The crude morbidity and mortality of a nationwide study were 8.4%–14.8% and 0.7%–1.6%, respectively, for surgical clipping for UCA and 4.0%–7.6% and 0.5%–0.7%, respectively, for endovascular coiling.^{2–4} A recent meta-analysis⁵ included 106,433 patients with 108,263 UCAs. Surgical treatment (54 studies) showed a pooled complication risk of 8.34% (range, 6.25%–11.10%) and a case-fatality rate of 0.10% (range, 0.00%–0.20%). Endovascular treatment (74 studies) showed a pooled clinical complication risk of 4.96% (95% confidence interval [CI], 4.00–6.12) and a case-fatality rate of 0.30% (95% CI, 0.20–0.40).

Key words

- Elderly patient
- Endovascular coiling
- Functional outcome
- Nationwide database
- Surgical clipping
- Unruptured cerebral aneurysm

Abbreviations and Acronyms

BI: Barthel Index

CI: Confidence interval

ICD-10: International Classification of Diseases, Tenth Revision

UCA: Unruptured cerebral aneurysms

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Citation: *World Neurosurg.* (2020) 136:e371–e379.

<https://doi.org/10.1016/j.wneu.2020.01.005>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

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However, almost all studies used biased data, and there have been few studies with propensity score analysis of reduced bias,^{6,7} and there has never been a propensity score analysis according to age-group.

This study aimed to clarify nonbiased morbidity at discharge and in-hospital mortality after surgical clipping and endovascular coiling for UCAs in nonelderly (<65 years) and elderly (≥65 years) patients, based on the Diagnosis Procedure Combination database, the national database of in-hospital patients in Japan.

METHODS

Ethical Approval

This study was approved by the institutional review board of the University of Hiroshima (number E-629) and the University of Tokyo (number 3501- [1]).

Informed Consent

Because of the anonymous nature of the data, the requirement for informed consent was waived.

Data Source

The Japanese Diagnosis Procedure Combination database is a national database of in-hospital patients in Japan, as described elsewhere.⁸⁻¹⁰ The database contains administrative claims and

abstract discharge data including the diagnoses and comorbidities on admission coded according to the *International Classification of Diseases, Tenth Revision (ICD-10)*; medical history; treatment methods; complications after the procedure; and Barthel Index (BI) at admission and discharge.

Patient and Aneurysm Selection and Data

This study included patients who were admitted to the hospital with a primary diagnosis of UCA from July 2010 to March 2015. Diagnoses of UCA were identified as ICD-10 code I671. This study included only patients at initial admission, only treatment with surgical clipping and/or endovascular coiling, and only single treatment of UCA, and excluded cases with unknown location, multiple aneurysms, no BI assessment, and other than 100 BI at admission. Consequently, this study included 15,671 patients with UCAs as described in another article,¹⁰ and the same data set was used in this study.

Propensity Score Matching Analysis

All 15,671 cases were divided into 2 groups: 8019 patients in the nonelderly group (<65 years) and 7649 patients in the elderly group (≥65 years). The background of the patients was biased originally in each group, and, therefore, we conducted propensity score matching analysis to reduce bias in the total, nonelderly, and elderly groups (Figure 1). However, because antiplatelet drugs were systematically administered before intervention in the

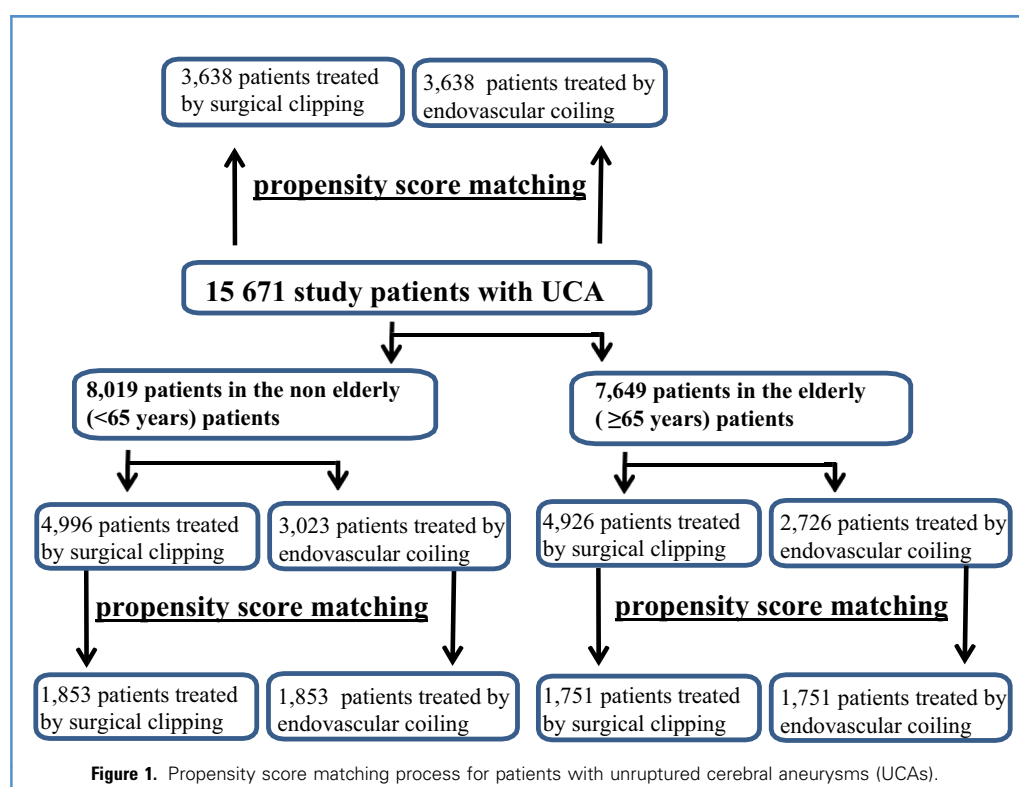


Figure 1. Propensity score matching process for patients with unruptured cerebral aneurysms (UCAs).

Table 1. Baseline Characteristics and Outcome of Unruptured Cerebral Aneurysm by Surgical Clipping and Endovascular Coiling

	Surgical Clipping (n = 9922)	Endovascular Coiling (n = 5749)	P Value
Age (years), median (interquartile range)	64.0 (57.0–70.0)	64.0 (54.0–71.0)	0.002*
Sex (male)	3107 (31.3)	1649 (28.7)	<0.001*
Location of aneurysms			<0.001*
Internal carotid artery	2614 (26.3)	3386 (58.9)	
Anterior communicating artery	1769 (17.8)	734 (12.8)	
Middle cerebral artery	4660 (47.0)	307 (5.3)	
Anterior cerebral artery	587 (5.9)	182 (3.2)	
Basilar artery	156 (1.6)	709 (12.3)	
Vertebral artery	126 (1.3)	381 (6.6)	
Other	10 (0.1)	50 (0.9)	
Hospital volume			<0.001*
1	3265 (32.9)	1857 (32.3)	
2	2957 (29.8)	2217 (38.6)	
3	3700 (37.3)	1675 (29.1)	
Academic (%)	2120 (21.4)	1743 (30.3)	<0.001*
Medical history			
Diabetes mellitus	940 (9.5)	450 (7.8)	<0.001*
Hypertension	4417 (44.5)	2015 (35.0)	<0.001*
Cerebral infarction	376 (3.8)	1234 (21.5)	<0.001*
Angina pectoris	341 (3.4)	239 (4.2)	0.021*
Myocardial infarction	5 (0.1)	7 (0.1)	0.12
Chronic heart disease	116 (1.2)	69 (1.2)	0.86
Hyperlipidemia	1880 (18.9)	957 (16.6)	<0.001*
Complication			
Intracerebral hemorrhage	27 (0.3)	15 (0.3)	0.9
Cerebral infarction	418 (4.2)	527 (9.2)	<0.001*
Subarachnoid hemorrhage	0 (0.0)	0 (0.0)	
Congestive heart failure	47 (0.5)	24 (0.4)	0.61
Pneumonia	172 (1.7)	104 (1.8)	0.73
Internal medicine on admission			
Antiplatelet	815 (8.2)	4902 (85.3)	<0.001*
Anticoagulation	178 (1.8)	117 (2.0)	0.28
Continues			

Table 1. Continued

	Surgical Clipping (n = 9922)	Endovascular Coiling (n = 5749)	P Value
Statin	923 (9.3)	561 (9.8)	0.35
Death (in-hospital mortality)	17 (0.2)	26 (0.5)	0.002*
Barthel Index at discharge			0.008*
Death	17 (0.2)	26 (0.5)	0.93
0	42 (0.4)	25 (0.4)	
5–30	39 (0.4)	23 (0.4)	
35–60	110 (1.1)	61 (1.1)	
65–85	187 (1.9)	96 (1.7)	
90	101 (1.0)	45 (0.8)	
95	159 (1.6)	66 (1.1)	
100	9267 (93.4)	5407 (94.1)	0.11
Values are number (%) except where indicated otherwise. *P < 0.05.			

patients with endovascular coiling, we excluded the valuable of antiplatelet drugs in the matching group based on too high an absolute standardized difference.

Statistical Analysis

Categorical variables were compared using the χ^2 test or Fisher exact test. Continuous variables were compared using the t test or Mann-Whitney U test, as appropriate. In addition, propensity score matching was used to balance the patients' backgrounds between the surgical clipping and endovascular coiling groups, except for antiplatelet drug. The propensity score was estimated using a logistic regression model for each treatment as a function of patients' age and sex, the location of the aneurysm, hospital volume, academic hospital, and medical history. C-statistics were calculated to test the goodness of fit. One-to-one matched analysis used the nearest-neighbor method without replacement with the closest estimated propensity score within a caliper (0.2 of the pooled standard deviation). The balance in baseline variables was estimated using standardized differences, where 10% was regarded as imbalanced. Clinical outcomes of poor BI <90 and in-hospital mortality were compared between the propensity score-matched groups according to surgical clipping and endovascular coiling in each age-group. All statistical analyses were performed with Stata version 15 software (Stata Corp LLC, College Station, Texas, USA).

Role of the Funding Source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Table 2. All Patient Characteristics in the Propensity Score—Matched Patients and Barthel Index at Discharge by Surgical Clipping and Endovascular Coiling

Variable	Matched Groups				P Value
	Surgical Clipping (n = 3638)	Endovascular Coiling (n = 3638)	Absolute Standardized Difference	Risk Difference, % (95% Confidence Interval)	
Age (years), mean	62.0	62.0	−0.6%		
Sex, male (%)	29.0	28.0	−2.9		
Location of aneurysms (%)					
Internal carotid artery	61.5	64.0	5.3		
Anterior communicating artery	18.1	17.2	−2.5		
Middle cerebral artery	7.8	8.4	2.5		
Anterior cerebral artery	4.6	4.6	−0.4		
Basilar artery	4.3	3.1	−6.6		
Vertebral artery	3.5	2.6	−5.0		
Other	0.3	0.1	−3.0		
Hospital volume (%)					
1	33.0	33.7	1.3		
2	33.5	32.5	−2.0		
3	33.5	33.8	0.7		
Academic (%)	25.0	23.0	−4.4		
Medical history (%)					
Diabetes mellitus	8.0	9.0	0.9		
Hypertension	38.0	38.0	0.9		
Cerebral infarction	7.0	6.0	−4.2		
Angina pectoris	4.0	4.0	1.4		
Myocardial infarction	0.0	0.0	1.0		
Chronic heart disease	1.0	1.0	−2.8		
Hyperlipidemia	17.0	18.0	2.0		
Internal medicine on admission					
Antiplatelet	—	—	—		
Anticoagulation	2.0	2.0	−2.1		
Statin	10.0	10.0	0.6		
Death (in-hospital mortality)	11 (0.3)	11 (0.3)			1.0
Barthel Index at discharge					0.048*
Death	11 (0.3)	11 (0.3)			
0	24 (0.7)	13 (0.4)			
5–30	17 (0.5)	12 (0.3)			
35–60	44 (1.2)	41 (1.1)			
65–85	82 (2.3)	65 (1.8)			
90	40 (1.1)	28 (0.8)			

Continues

Table 2. Continued

Variable	Matched Groups				P Value
	Surgical Clipping (n = 3638)	Endovascular Coiling (n = 3638)	Absolute Standardized Difference	Risk Difference, % (95% Confidence Interval)	
95	63 (1.7)	40 (1.1)			
100	3357 (92.3)	3428 (94.2)			0.028*
Death	178 (4.9)	142 (3.9)		−1.0 (−3.6 to 2.3)	0.040*
0					
5–30					
35–60					
65–85					
90	3460 (95.1)	3496 (96.1)			
95					
100					

*P < 0.05.

RESULTS

Table 1 shows the baseline characteristics and outcome of UCAs by surgical clipping and endovascular coiling. Almost all variables were significantly different in backgrounds.

Table 2 shows the patient characteristics in the propensity score-matched analysis of total patients of BI at discharge after surgical clipping and endovascular coiling. Baseline populations included 9922 cases in the surgical clipping group and 3638 cases in the endovascular coiling group. Propensity score-matched analysis compared 3638 cases in each group. In the unmatched groups, absolute standardized differences >10% were seen according to aneurysm location, hospital volume, rate of an academic institute, hypertension, cerebral infarction, antiplatelet drug, and statin drug. After propensity score matching excluding antiplatelet drug, the baseline characteristics of the patients were well balanced between the groups; absolute standardized differences were within 10% for all variables. The C statistic for goodness of fit was 0.78 (95% CI, 0.77–0.79). The proportions of in-hospital mortality in the propensity score-matched groups of total patients showed no significant difference ($P = 1.0$) between surgical clipping and endovascular coiling. However, total patients treated with surgical clipping compared with endovascular coiling were significantly associated with a higher rate of the morbidity in BI <90 at discharge (4.9% vs. 3.9%; $P = 0.040$; risk difference, −1.0%; 95% CI, −3.6 to 2.3%).

Table 3 shows the patient characteristics in the propensity score-matched analysis of the nonelderly patients of BI at discharge after surgical clipping and endovascular coiling. Baseline populations included 4996 cases in the surgical clipping group and 3023 cases in the endovascular coiling group. Propensity score-matched analysis compared 1853 cases in each group. In the unmatched groups, absolute standardized differences >10% were seen according to age, aneurysm location, hospital volume, rate of an academic

institute, hypertension, cerebral infarction, and antiplatelet drug. After propensity score matching excluding antiplatelet drug, the baseline characteristics of the patients were well balanced between the groups; absolute standardized differences were within 10% for all variables. The C statistic for goodness of fit was 0.76 (95% CI, 0.75–0.77). The proportions of BI <90 at discharge (2.4% vs. 2.6%; $P = 0.67$; risk difference, 0.22%; 95% CI, −0.79 to 1.22%), and in-hospital mortality ($P = 1.0$) in the propensity score-matched groups of nonelderly patients showed no significant difference between surgical clipping and endovascular coiling.

Table 4 shows the patient characteristics in the propensity score-matched analysis of the elderly patients of BI at discharge by surgical clipping and endovascular coiling. Baseline populations included 4926 cases in the surgical clipping group and 2726 cases in the endovascular coiling group. Propensity score-matched analysis compared 1751 cases in each group. After propensity score matching excluding antiplatelet drug, the baseline characteristics of the patients were well balanced between the groups; absolute standardized differences were within 10% for all variables. The C statistic for goodness of fit was 0.75 (95% CI, 0.74–0.75). The proportions of in-hospital mortality in the propensity score-matched groups of elderly patients showed no significant difference ($P = 0.56$) between surgical clipping and endovascular coiling. However, elderly patients treated with surgical clipping compared with endovascular coiling were significantly associated with a higher rate of morbidity in BI <90 at discharge (8.1% vs. 5.0%; $P < 0.001$; risk difference, −3.1%; 95% CI, −4.8% to 1.5%).

DISCUSSION

Many studies worldwide have reported better outcome after endovascular coiling than after surgical clipping.^{2–4} According to some studies with nonbiased propensity score matching analysis,^{6,7}

Table 3. Nonelderly Patients (<65 years old) Characteristics in the Propensity Score–Matched Patients and Barthel Index at Discharge by Surgical Clipping and Endovascular Coiling

Variable	Matched Groups				P Value
	Surgical Clipping (n = 1853)	Endovascular Coiling (n = 1853)	Absolute Standardized Difference (%)	Risk Difference, % (95% Confidence Interval)	
Age (years), mean	53.2	53.4	2.2		
Sex, male (%)	32.0	31.0	−0.8		
Location of aneurysms (%)					
Internal carotid artery	64.8	67.4	5.6		
Anterior communicating artery	15.5	15.3	−0.4		
Middle cerebral artery	7.4	7.9	1.6		
Anterior cerebral artery	4.2	3.7	−2.5		
Basilar artery	4.4	3.3	−5.9		
Vertebral artery	3.4	2.1	−7.9		
Other	0.3	0.3	0.0		
Hospital volume (%)					
1	31.0	31.2	0.6		
2	32.9	32.6	−0.6		
3	36.1	36.1	0.0		
Academic (%)	27.0	25.0	−3.1		
Medical history (%)					
Diabetes mellitus	6.0	6.0	0.4		
Hypertension	32.0	32.0	1.8		
Cerebral infarction	6.0	5.0	−5.8		
Angina pectoris	2.0	3.0	1.0		
Myocardial infarction	0.0	0.0	1.9		
Chronic heart disease	1.0	1.0	1.3		
Hyperlipidemia	14.0	14.0	0.3		
Internal medicine on admission (%)					
Antiplatelet	—	—	—		
Anticoagulation	1.0	1.0	−3.6		
Statin	7.0	8.0	0.2		
Death (in-hospital mortality)	4 (0.2)	4 (0.2)			1.0
Barthel Index at discharge					0.75
Death	4 (0.2)	4 (0.2)			
0	7 (0.4)	6 (0.3)			
5–30	4 (0.2)	2 (0.1)			
35–60	8 (0.4)	12 (0.7)			
65–85	21 (1.1)	24 (1.3)			
90	13 (0.7)	8 (0.4)			

Continues

Table 3. Continued

Variable	Matched Groups				P Value
	Surgical Clipping (n = 1853)	Endovascular Coiling (n = 1853)	Absolute Standardized Difference (%)	Risk Difference, % (95% Confidence Interval)	
95	17 (0.9)	11 (0.6)			
100	1779 (96.0)	1786 (96.4)			0.61
Death	44 (2.4)	48 (2.6)		0.22 (−0.79 to 1.22)	0.67
0					
5–30					
35–60					
65–85					
90	1809 (97.6)	1805 (97.4)			
95					
100					

surgical clipping was associated with similar mortality risk but significantly higher morbidity risk compared with endovascular coiling, as in the total cases in our study. In addition, we analyzed nonelderly and elderly groups. As a result, no significantly different functional outcomes at discharge and in-hospital mortality between surgical clipping and endovascular coiling for UCA in nonelderly patients were confirmed by propensity score-matched analysis in Japan. This result may show the advantage of surgical clipping for nonelderly patients, because 41% of aneurysms showed recurrence, of which 26% required retreatment according to a long-term follow-up study of endovascular coiling.¹¹ On the other hand, a recent meta-analysis⁵ showed that the pooled crude risk of procedural clinical complications in method of outcome assessment according to ICD-coded databases was 8.91 (range, 6.38–12.31), twice that of medical records of 4.42 (range, 3.49–5.59), in endovascular treatment, and the risk of ICD-coded administrative databases was 20.38 (range, 14.69–27.56), 3 times that of medical records of 6.43 (range, 4.69–8.75), in neurosurgical treatment. However, our results of the nationwide database by ICD-coded were the same as those of a recent meta-analysis.⁵ Because the same meta-analysis⁵ showed that the risk for endovascular treatment in Asia was 3.31 (95% CI, 2.46–4.44), about half that of other areas, and 3.34 (95% CI, 2.88–6.50) for neurosurgical treatment, about one fifth that of other areas. Therefore, such procedures might be safer in Asia, especially neurosurgical treatment, compared with other areas. In Japan, the proportion of surgical clipping seemed higher than in other countries. Several possible reasons can be speculated for this phenomenon, including better outcome of surgical clipping, historically more surgical clipping than endovascular coiling, shortage of endovascular doctors, discussion about both treatments by neurosurgeons, >9500 available neurosurgeons and >7500 board-certified neurosurgeons, under the universal health care system, and so on, in Japan.¹² However, convenience, quicker recovery, and less invasiveness have recently been reported as attractive to patients worldwide.

For elderly patients with UCA, several scoring studies may support the decision making.^{13,14,15} If the risk of rupture is higher than in observationally conservative management, we interventional treatment should be selected and endovascular coiling should be considered for elderly patients with UCA.

Limitations

There are several limitations to this study using an administrative database.^{16,17} First, only in-hospital data are included in this database and no information after discharge is available. Second, planned prospective cohorts or registries were not conducted in this study. Therefore, data are less well validated and less detailed in this database. Third, this database did not include the aneurysm size, morphology, history of subarachnoid hemorrhage, and family history of subarachnoid hemorrhage. These factors were potentially unmeasured confounders in this database, as in other nationwide databases. Fourth, the propensity score-matched analysis was not complete in this study, because the antiplatelet drug was intentionally administered for endovascular coiling. If we include this valuable in the propensity score-matched analysis, few patients were residual in the matched cohort. Fifth, the general anesthesia invasion was not evaluated in this study. Because all patients receiving surgical clipping underwent under general anesthesia, and patients receiving endovascular coiling were treated under both general anesthesia and local anesthesia, the minimal invasiveness of surgical clipping does not reach that of endovascular coiling.

CONCLUSIONS

In elderly patients with UCA, a better outcome at discharge was found after endovascular coiling. However, no significantly different functional outcome at discharge between surgical clipping and endovascular coiling for UCA in nonelderly patients was confirmed by propensity score-matched analysis from a nationwide database in Japan.

Table 4. Elderly Patient (≥ 65 years old) Characteristics in the Propensity Score–Matched Patients and Barthel Index at Discharge by Surgical Clipping and Endovascular Coiling

Variable	Matched Groups		Absolute Standardized Difference (%)	Risk Difference, % (95% Confidence Interval)	P Value
	Surgical Clipping (n = 1751)	Endovascular Coiling (n = 1751)			
Age (years), mean	71.6	71.5	−2.5		
Sex, male (%)	23.0	22.0	−0.5		
Location of aneurysms					
Internal carotid artery	58.0	58.0	−0.1		
Anterior communicating artery	19.6	20.2	1.6		
Middle cerebral artery	9.0	9.2	0.8		
Anterior cerebral artery	5.5	5.6	0.5		
Basilar artery	4.2	2.9	−7.4		
Vertebral artery	3.4	3.9	2.4		
Other	0.3	0.3	0.0		
Hospital volume					
1	35.0	36.2	2.6		
2	33.7	32.8	−1.8		
3	31.4	31.0	−0.9		
Academic (%)	23.0	22.0	−0.5		
Medical history (%)					
Diabetes mellitus	10.0	11.0	5.4		
Hypertension	44.0	45.0	2.6		
Cerebral infarction	9.0	8.0	−4.3		
Angina pectoris	5.0	6.0	4.3		
Myocardial infarction	0.0	0.0	0.0		
Chronic heart disease	2.0	2.0	−0.4		
Hyperlipidemia	20.0	21.0	3.4		
Internal medicine on admission (%)					
Antiplatelet	—	—	—		
Anticoagulation	3.0	3.0	−1.3		
Statin	13.0	12.0	−2.3		
Death (in-hospital mortality)	7 (0.4)	5 (0.3)			0.56
Barthel Index at discharge					0.008*
Death	7 (0.4)	5 (0.3)			
0	19 (1.1)	8 (0.5)			
5–30	14 (0.8)	10 (0.6)			
35–60	38 (2.2)	21 (1.2)			
65–85	64 (3.7)	43 (2.5)			
90	28 (1.6)	21 (1.2)			

Continues

Table 4. Continued

Variable	Matched Groups		Absolute Standardized Difference (%)	Risk Difference, % (95% Confidence Interval)	P Value
	Surgical Clipping (n = 1751)	Endovascular Coiling (n = 1751)			
95	43 (2.5)	31 (1.8)			
100	1538 (87.8)	1612 (92.1)			<0.001*
Death	142 (8.1)	87 (5.0)		−3.1 (−4.8 to −1.5)	<0.001*
0					
5–30					
35–60					
65–85					
90	1609 (91.9)	1664 (95.0)			
95					
100					

*P < 0.05.

REFERENCES

- Jalbert JJ, Isaacs AJ, Kamel H, Sedrakyan A. Clipping and coiling of unruptured intracranial aneurysms among Medicare beneficiaries, 2000 to 2010. *Stroke*. 2015;46:2452–2457.
- Alshekhlee A, Mehta S, Edgell RC, et al. Hospital mortality and complications of electively clipped or coiled unruptured intracranial aneurysm. *Stroke*. 2010;41:1471–1476.
- Brinjikji W, Rabinstein AA, Nasr DM, Lanzino G, Kallmes DF, Cloft HJ. Better outcomes with treatment by coiling relative to clipping of unruptured intracranial aneurysms in the United States, 2001–2008. *AJNR Am J Neuroradiol*. 2011;32:1071–1075.
- Silva NA, Shao B, Sylvester MJ, Eloy JA, Gandhi CD. Unruptured aneurysms in the elderly: perioperative outcomes and cost analysis of endovascular coiling and surgical clipping. *Neurosurg Focus*. 2018;44:E4.
- Algra AM, Lindgren A, Vergouwens MDI, et al. Procedural clinical complications, case-fatality risks, and risk factors in endovascular and neurosurgical treatment of unruptured intracranial aneurysms: a systematic review and meta-analysis. *JAMA Neurol*. 2019;76:282–293.
- McDonald JS, McDonald RJ, Fan J, Kallmes DF, Lanzino G, Cloft HJ. Comparative effectiveness of unruptured cerebral aneurysm therapies: propensity score analysis of clipping versus coiling. *Stroke*. 2013;44:988–994.
- McDonald JS, McDonald RJ, Fan J, Kallmes DF, Lanzino G, Cloft HJ. Comparative effectiveness of ruptured cerebral aneurysm therapies: propensity score analysis of clipping versus coiling. *AJNR Am J Neuroradiol*. 2014;35:164–169.
- Yasunaga HM, Horiguchi H, Fushimi K, Matsuda S. Clinical epidemiology and health services research using the Diagnosis Procedure Combination Database in Japan. *Asian Pac J Dis Manage*. 2013;7:19–24.
- Ikawa F, Michihata N, Matsushige T, et al. In-hospital mortality and poor outcome after surgical clipping and endovascular coiling for aneurysmal subarachnoid hemorrhage using nationwide databases: a systematic review and meta-analysis [e-pub ahead of print]. *Neurosurg Rev*. <https://doi.org/10.1007/s10143-019-01096-2>, accessed April 2, 2019.
- Ikawa F, Michihata N, Akiyama Y, et al. Treatment risk for elderly patients with unruptured cerebral aneurysm from a nationwide database in Japan. *World Neurosurg*. 2019;132:e89–e98.
- Chalouhi N, Bovenzi CD, Thakkar V, et al. Long-term catheter angiography after aneurysm coil therapy: results of 209 patients and predictors of delayed recurrence and retreatment. *J Neurosurg*. 2014;121:1102–1106.
- Ikawa F, Hidaka T, Yoshiyama M, et al. Characteristics of cerebral aneurysms in Japan. *Neurol Med Chir (Tokyo)*. 2019;59:399–406.
- Greving JP, Wermer MJ, Brown RD Jr, et al. Development of the PHASES score for prediction of risk of rupture of intracranial aneurysms: a pooled analysis of six prospective cohort studies. *Lancet Neurol*. 2014;13:59–66.
- Backes D, Rinkel GJE, Greving JP, et al. ELAPSS score for prediction of risk of growth of unruptured intracranial aneurysms. *Neurology*. 2017;88:1600–1606.
- Tominari S, Morita A, Ishibashi T, et al. Prediction model for 3-year rupture risk of unruptured cerebral aneurysms in Japanese patients. *Ann Neurol*. 2015;77:1050–1059.
- Kestle JR. Administrative database research. *J Neurosurg*. 2015;122:441–442.
- Oravec CS, Motiwala M, Reed K, et al. Big Data research in neurosurgery: a critical look at this popular new study design. *Neurosurgery*. 2018;82:728–746.

Conflict of interest statement: This work was supported by the Japan Society for the Promotion of Science, Grant-in-Aid for Scientific Research (C) 17K10829.

Received 30 October 2019; accepted 2 January 2020

Citation: *World Neurosurg*. (2020) 136:e371–e379.

<https://doi.org/10.1016/j.wneu.2020.01.005>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

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